

Math 241, F04  
Quiz 10

Name:  
SSN: xxx-xx-

**Instructions:** There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

**Problem 1.** (10 points) Let

$$\mathbf{F}(x, y, z) = x^2\mathbf{i} - 2xy\mathbf{j} + yz^2\mathbf{k}$$

a) Find  $\text{div } \mathbf{F}$ . (Hint:  $\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$ )

**Solution:** Let  $M = x^2$ ,  $N = -2xy$ ,  $P = yz^2$ , then

$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z} = 2x - 2x + 2yz = 2yz$$

b) Find  $\text{curl } \mathbf{F}$ . (Hint:  $\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$ )

**Solution:**

$$\begin{aligned}\text{curl } \mathbf{F} = \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & P \end{vmatrix} \\ &= \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z}\right)\mathbf{i} + \left(\frac{\partial M}{\partial z} - \frac{\partial P}{\partial x}\right)\mathbf{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)\mathbf{k} \\ &= (z^2 - 0)\mathbf{i} + (0 - 0)\mathbf{j} + (-2y - 0)\mathbf{k} \\ &= z^2\mathbf{i} - 2y\mathbf{k}\end{aligned}$$

**Problem 2.** (10 points) Find the work done by the force field  $\mathbf{F}$  in moving a particle along the curve  $C$  where  $\mathbf{F}(x, y) = e^x \mathbf{i} - e^{-y} \mathbf{j}$  and  $C$  is the curve  $x = 3 \ln t$ ,  $y = \ln 2t$ ,  $1 \leq t \leq 5$ .

**Solution:** Let  $M = e^x$  and  $N = -e^{-y}$ , then the work done by  $\mathbf{F}$  is

$$\begin{aligned}
 W &= \int_C \mathbf{F} \cdot d\mathbf{r} \\
 &= \int_C M \, dx + N \, dy \\
 &= \int_C e^x \, dx - e^{-y} \, dy \\
 &= \int_1^5 e^{3 \ln t} \cdot \left(\frac{3}{t}\right) dt - \int_1^5 e^{-\ln 2t} \cdot \left(\frac{1}{2t}\right) dt \\
 &= \int_1^5 \left(3t^2 - \frac{1}{4t^2}\right) dt \\
 &= \left[t^3 + \frac{1}{4t}\right]_1^5 \\
 &= (125 + 1/20) - (1 + 1/4) \\
 &= 123.8
 \end{aligned}$$